

# SESSION 6 ASSIGNMENT

## Employment Coaching and Recruitment Workshop One Competency Interview Practice

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| <b>Selected Competency</b> | Problem Solving                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Interview Question</b>  | Can you describe a time when you faced a complex technical problem during a project and how you went about solving it?                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Situation</b>           | During my training in the DEPI React track, we were working on a team project to build an E-commerce web application. Towards the end of the project, we noticed a significant performance issue; the product listing page was rendering very slowly and sometimes lagging when users typed in the search bar.                                                                                                                                                                                                                            |
| <b>Task</b>                | As a front-end developer on the team, my task was to identify the root cause of this performance issue and fix it before our final project submission deadline, which was only two days away.                                                                                                                                                                                                                                                                                                                                             |
| <b>Action</b>              | First, I used Chrome Developer Tools (React Profiler) to debug the application. I discovered that the issue was caused by unnecessary re-renders of the entire product list component every time a single keystroke was entered in the search bar. To solve this, I applied the <code>useMemo</code> hook to memoize the product list so it only re-renders when the actual data changes. Additionally, I implemented a "debounce" custom hook for the search input to delay the API calls and state updates until the user stops typing. |

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| <b>Result</b>                              | <p>As a result of these optimizations, the unnecessary re-renders were eliminated, and the page performance improved by 70%. The search feature became perfectly smooth, and we successfully submitted a highly optimized project on time, receiving positive feedback from our instructors.</p>                |
| <b>One Improvement Made After Feedback</b> | <p>After reviewing my initial answer, I improved the "Result" section by adding a specific measurable metric ("improved by 70%") instead of just saying "it became faster", and I made sure to focus on my specific technical actions (useMemo and debounce) rather than talking in general about the team.</p> |